

LAB 1
Digital Systems and Microcontrollers (DSM)
International Institute of Information Technology, Hyderabad

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1 Experiment 1

1.1 Aim

Familiarization with Digital Test Kit and use it to check working of NOT gate.

1.2 Components & Equipment Required

- Digital Test Kit (includes power supply, switches, LEDs and breadboard)
- IC 7404 (TTL hex inverter)
- Connecting Wires

1.3 Procedure

1. Digital Test Kit Setup & Verification:

- (a) Test the functionality of the input toggle switches (S1–S12) and dual-color LED display indicators by applying direct inputs to verify logic HIGH (5V) and LOW (0V) states.

2. IC 7404 Hex Inverter Wiring & Logic Verification:

- (a) Mount the IC 7404 across the center notch of the breadboard.
- (b) Connect Pin 14 (V_{CC}) to the +5V power rail and Pin 7 (GND) to the ground rail.
- (c) Wire an input pin to one of the digital input switches (S1) on the test kit.
- (d) Wire the corresponding inverter output pin to an LED display terminal on the test kit.
- (e) Toggle input switch S1 between HIGH and LOW logic levels, observing and recording the active LED indicators and output logic states.

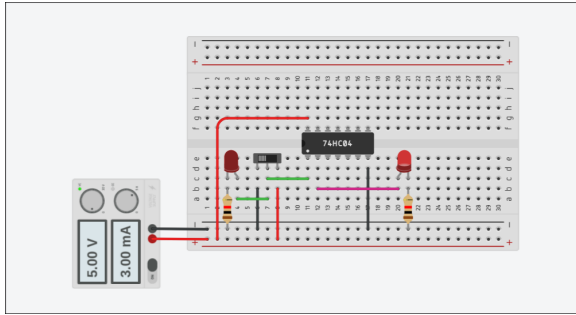
1.4 Observations

Table 1: Observations for Experiment 1

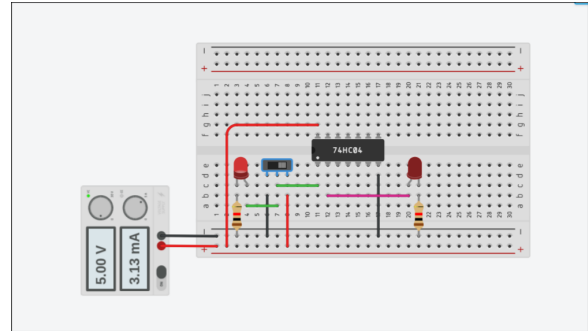
Switch State	LED Indicator
OFF (LOW)	Red (HIGH)
ON (HIGH)	Green (LOW)

1.5 Tinkercad Simulation

Tinkercad Link: [Click here](#)



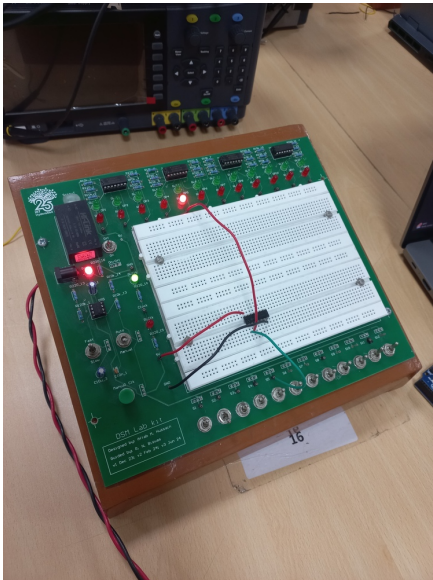
(a) Switch OFF (Output HIGH)



(b) Switch ON (Output LOW)

Figure 1: Tinkercad Simulation Screenshot

1.6 Hardware Setup Photograph



(a) Switch OFF (Output HIGH / Green LED) (b) Switch ON (Output LOW / Green LED)

Figure 2: Hardware Setup

1.7 Conclusion

- **Digital Test Kit Functional Verification**
- **IC 7404 Operation:** The IC 7404 TTL Hex Inverter behaved according to boolean logic rules ($Y = \bar{A}$) as expected.
- **Experimental Validation:** The experimental observations matched the theoretical truth table of NOT gate.

2 Experiment 2

2.1 Aim

- Learn how to write and compile code for ATmega328P microcontroller in Arduino.
- Write a program output "ON" and "OFF" to serial monitor based on output of NOT gate.

2.2 Components & Equipment Required

- Digital Test Kit (includes power supply, switches, LEDs and breadboard)
- IC 7404 (TTL hex inverter)
- Connecting Wires
- Arduino Uno
- USB A-to-B cable

2.3 Procedure

1. Hardware Setup:

- (a) Retain the IC 7404 setup on the breadboard from Experiment 1.
- (b) Connect a common ground wire between the ground rail (GND) of the Digital Test Kit and a GND pin on the Arduino Uno to ensure a unified logic reference level.
- (c) Tap the output signal of the inverter using a jumper wire and connect it directly to a digital input pin on the Arduino Uno board.

2. Software:

- (a) Download and install the Arduino IDE.
- (b) Connect the Arduino Uno to the computer via a USB A-to-B cable.
- (c) Write the code to continuously read the digital voltage level using `digitalRead()` and output the corresponding logic state to the Serial Monitor.
- (d) Select the correct board (*Arduino Uno*) and Serial Port under the **Tools** menu. Then compile and upload the sketch to the microcontroller.

3. Observation:

- (a) Open the *Serial Monitor* inside the Arduino IDE.
- (b) Toggle the physical switch between LOW and HIGH logic levels.
- (c) Simultaneously observe and cross-verify:
 - The position of the input toggle switch.
 - The active state of the dual-color indicator LED.
 - The printed logic state on the Arduino IDE Serial Monitor.
- (d) Verify that the state logged on the Serial Monitor matches the inverted logic output displayed by the indicator LED.

2.4 Arduino Code

```
1 #define IN_PIN 8
2
3 void setup() {
4     Serial.begin(9600);
5     pinMode(IN_PIN, INPUT);
6 }
7
8 void loop() {
9     int val = digitalRead(IN_PIN);
10
11     if (val == HIGH) Serial.println("ON");
12     else Serial.println("OFF");
13
14     delay(500);
15 }
```

Listing 1: Arduino Code for Experiment 2

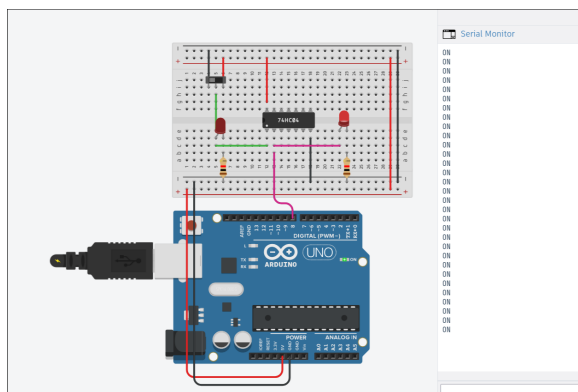
2.5 Observations

Table 2: Observations for Experiment 2

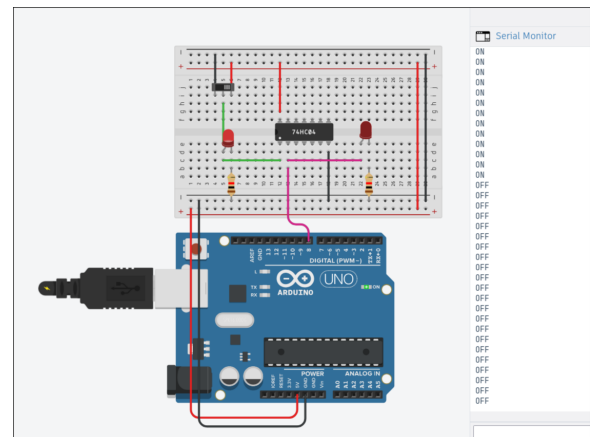
Switch	Indicator LED	Serial Monitor Output
OFF	Red (Logic 1)	ON
ON	Green (Logic 0)	OFF

2.6 Tinkercad Simulation

Tinkercad Link: [Click here](#)



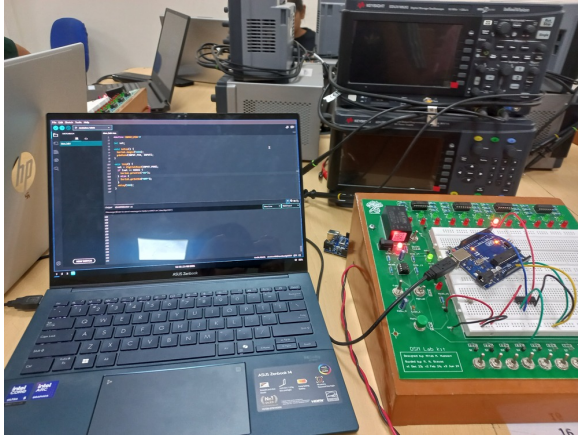
(a) Switch OFF



(b) Switch ON

Figure 3: Tinkercad Simulation Screenshot

2.7 Hardware Setup Photograph



(a) Switch OFF



(b) Switch ON

Figure 4: Hardware Setup

2.8 Conclusion

We successfully configured the software environment to write, compile, and upload C++ code to the Arduino Uno. The real-time readings captured on the Serial Monitor matched with the physical LED indicators and input switch states.